

ECE238L Computer Logic Design

Lab 0: Design and Simulate Gates

Section 003: Wednesday 2:00-4:45

Name: Tomas Salaz

Teammate: Nathan Nail (Only Part A)

Lab 0 Part I:

VHDL Source Code, Simulation Code (Test Bench) and Constraints

Lab_0_tb.vhd

C:/Users/salaz25/ECE238/Lab_0/Lab_0.srcs/sim_1/new/Lab_0_tb.vhd

```
17 -- Additional Comments:
18 --
19 -----
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using...
26
27
28 entity Lab_0_tb is
29   -- Port ( );
30 end Lab_0_tb;
31
32
33 architecture bench of Lab_0_tb is
34   component Lab_0
35     Port (A,B,C : in STD_LOGIC;
36           F: out STD_LOGIC);
37   end component;
38
39   signal A_tb, B_tb, C_tb: STD_LOGIC;
40   signal F_tb : STD_LOGIC;
41
42 begin
43
44 uut: Lab_0 port map (A => A_tb,
45   B => B_tb,
46   C => C_tb,
47   F => F_tb );
48
49 stimulus: process
50 begin
51   A_tb <= '0'; B_tb <= '0'; C_tb <= '1'; wait for 100 ns;
52   A_tb <= '0'; B_tb <= '1'; C_tb <= '0'; wait for 100 ns;
53   A_tb <= '0'; B_tb <= '1'; C_tb <= '1'; wait for 100 ns;
54   A_tb <= '0'; B_tb <= '0'; C_tb <= '0'; wait for 100 ns;
55   A_tb <= '1'; B_tb <= '0'; C_tb <= '1'; wait for 100 ns;
56   A_tb <= '1'; B_tb <= '1'; C_tb <= '0'; wait for 100 ns;
57   A_tb <= '1'; B_tb <= '1'; C_tb <= '1'; wait for 100 ns;
58   A_tb <= '1'; B_tb <= '0'; C_tb <= '0'; wait for 100 ns;
59   wait; -- default wait
60 end process;
61
62 end bench;
```

Lab_0.vhd

C:/Users/salaz25/ECE238/Lab_0/Lab_0.srcs/sources_1/new/Lab_0.vhd

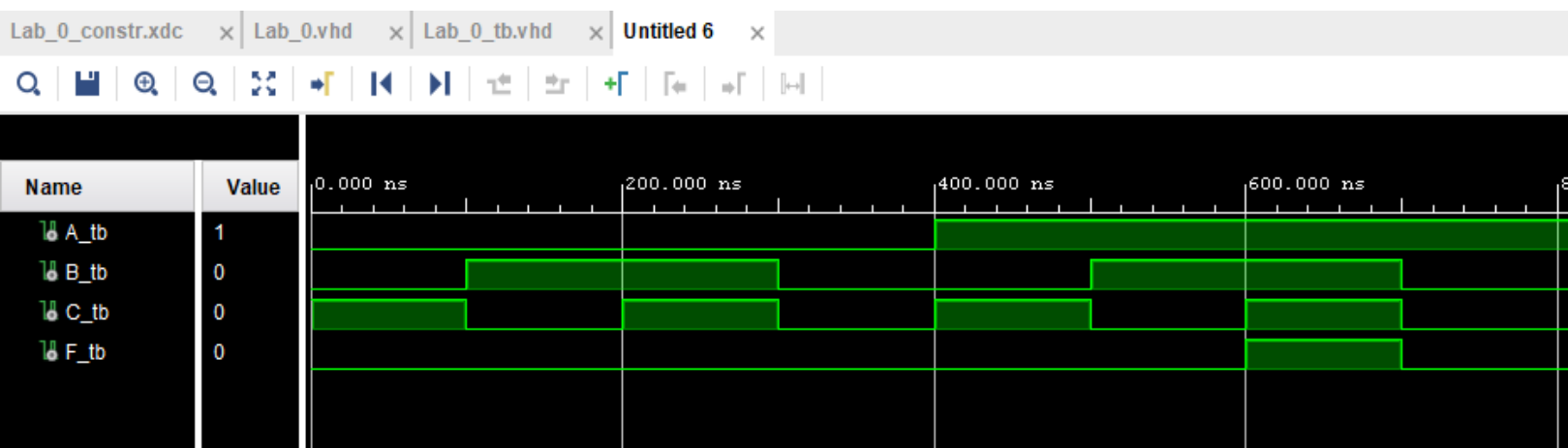
```
1
2 -- Company: UNM
3 -- Engineer: NATHAN NAIL, TOMAS SALAZ
4 --
5 -- Create Date: 01/28/2025 02:25:57 PM
6 -- Design Name:
7 -- Module Name: Lab_0 - Boolean_Funciton
8 -- Project Name:
9 -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using...
26
27
28 entity Lab_0 is
29   Port ( A, B, C : in STD_LOGIC;
30         F : out STD_LOGIC);
31 end Lab_0;
32
33
34 architecture Boolean_Funciton of Lab_0 is
35 begin
36   F <= A and B and C; --establish out logic
37
38 end Boolean_Funciton;
```

Lab_0_constr.xdc

C:/Users/salaz25/ECE238/Lab_0/Lab_0.srcs/constrs_1/new/Lab_0_constr.xdc

```
1 set_property PACKAGE_PIN J15 [get_ports A]
2 set_property IOSTANDARD LVCMOS33 [get_ports B]
3 set_property IOSTANDARD LVCMOS33 [get_ports C]
4 set_property IOSTANDARD LVCMOS33 [get_ports F]
5 set_property IOSTANDARD LVCMOS33 [get_ports A]
6 set_property PACKAGE_PIN L16 [get_ports B]
7 set_property PACKAGE_PIN M13 [get_ports C]
8 set_property PACKAGE_PIN H17 [get_ports F]
9
```

Wave Form



Schematic, Truth Table, and Wave Form

Schematic

A

I

A_IBUF_inst

IBUF

B

I

B_IBUF_inst

IBUF

C

I

C_IBUF_inst

IBUF

F_OBUF_inst_i_1

LUT3

I0

I1

I2

O

F_OBUF_inst

OBUF

F

Cell Properties

F_OBUF_inst_i_1

I2

I1

I0

O=I0 & I1 & I2

0

0

0

0

0

0

1

0

0

1

0

0

0

1

1

0

1

0

0

0

1

0

1

0

1

1

0

0

1

1

1

1

Edit LUT Equation...

ver

Nets

Cell Pins

Truth Table

VHDL Source Code, Simulation Code (Test Bench), and Constraints

Wave Form

Name	Value
A_tb	1
B_tb	0
F_NOT_tb	0
F_AND_tb	0
F_OR_tb	1

The timing diagram shows five digital signals over a 300 ns period. The signals are: A_tb (green, high), B_tb (blue, low), F_NOT_tb (green, high), F_AND_tb (blue, low), and F_OR_tb (green, high). The signals are constant over time.

Truth Table and Schematic

SourcesNetlist? _ □ □ 0

Design Sources (1)

Lab_0A(Boolean_Function) (Lab_0A.vhd)

Constraints (1)

constrs_1 (1)

Lab_0A_cstr.xdc (target)

Simulation Sources (1)

sim_1 (1)

HierarchyLibrariesCompile Order

Cell Properties? _ □ □ ×

F_OR_OBUF_inst_i_1

I1	I0	O=I0 + I1
0	0	0
0	1	1
1	0	1
1	1	1

Edit LUT Equation...

GeneralPropertiesPowerNetsCell PinsTruth Table

Project SummaryLab_0A.vhdLab0A_tb.vhdLab_0A_cstr.xdcSchematic? _ □ □ 0

8 Cells5 I/O Ports10 Nets

```
graph LR; A[A] --> A_IBUF[A_IBUF_inst IBUF]; B[B] --> B_IBUF[B_IBUF_inst IBUF]; A_IBUF --> LUT1[F_NOT_OBUF_inst_i_1 LUT1]; B_IBUF --> LUT1; A_IBUF --> LUT2[F_AND_OBUF_inst_i_1 LUT2]; B_IBUF --> LUT2; A_IBUF --> LUT2; B_IBUF --> LUT2; LUT1 --> F_NOT_OBUF[F_NOT_OBUF_inst OBUF]; LUT2 --> F_AND_OBUF[F_AND_OBUF_inst OBUF]; LUT2 --> F_OR_OBUF[F_OR_OBUF_inst OBUF]; F_NOT_OBUF --> F_NOT[F_NOT]; F_AND_OBUF --> F_AND[F_AND]; F_OR_OBUF --> F_OR[F_OR];
```

Lab 0 Part B:

VHDL Source Code, Simulation Code (Test Bench) and Constraints

```
OB.vhd
-- Company:
-- Engineer:
-- Create Date: 02/04/2025 01:39:01 PM
-- Design Name:
-- Module Name: Lab_0B - Boolean_Function
-- Project Name:
-- Target Devices:
-- Tool Versions:
-- Description:
-- Dependencies:
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity Lab_0B is
    Port ( A, B : in STD_LOGIC;
          F_NAND : out STD_LOGIC;
          F_NOR : out STD_LOGIC;
          F_XOR : out STD_LOGIC;
          F_XNOR : out STD_LOGIC;
          F_BUFF : out STD_LOGIC);
end Lab_0B;

architecture Boolean_Function of Lab_0B is
begin
    F_NAND <= not (A and B);
    F_NOR <= not (A or B);
    F_XOR <= A xor B;
    F_XNOR <= A xnor B;
    F_BUFF <= A;
end Boolean_Function;

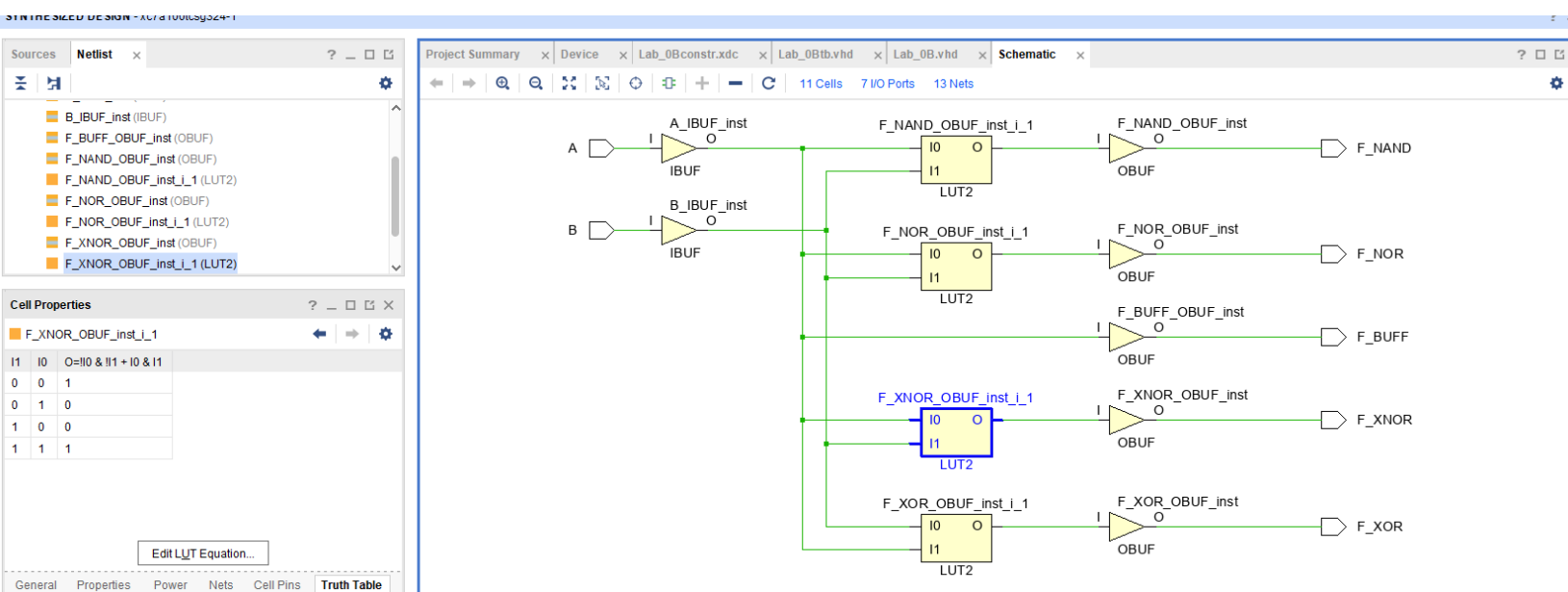
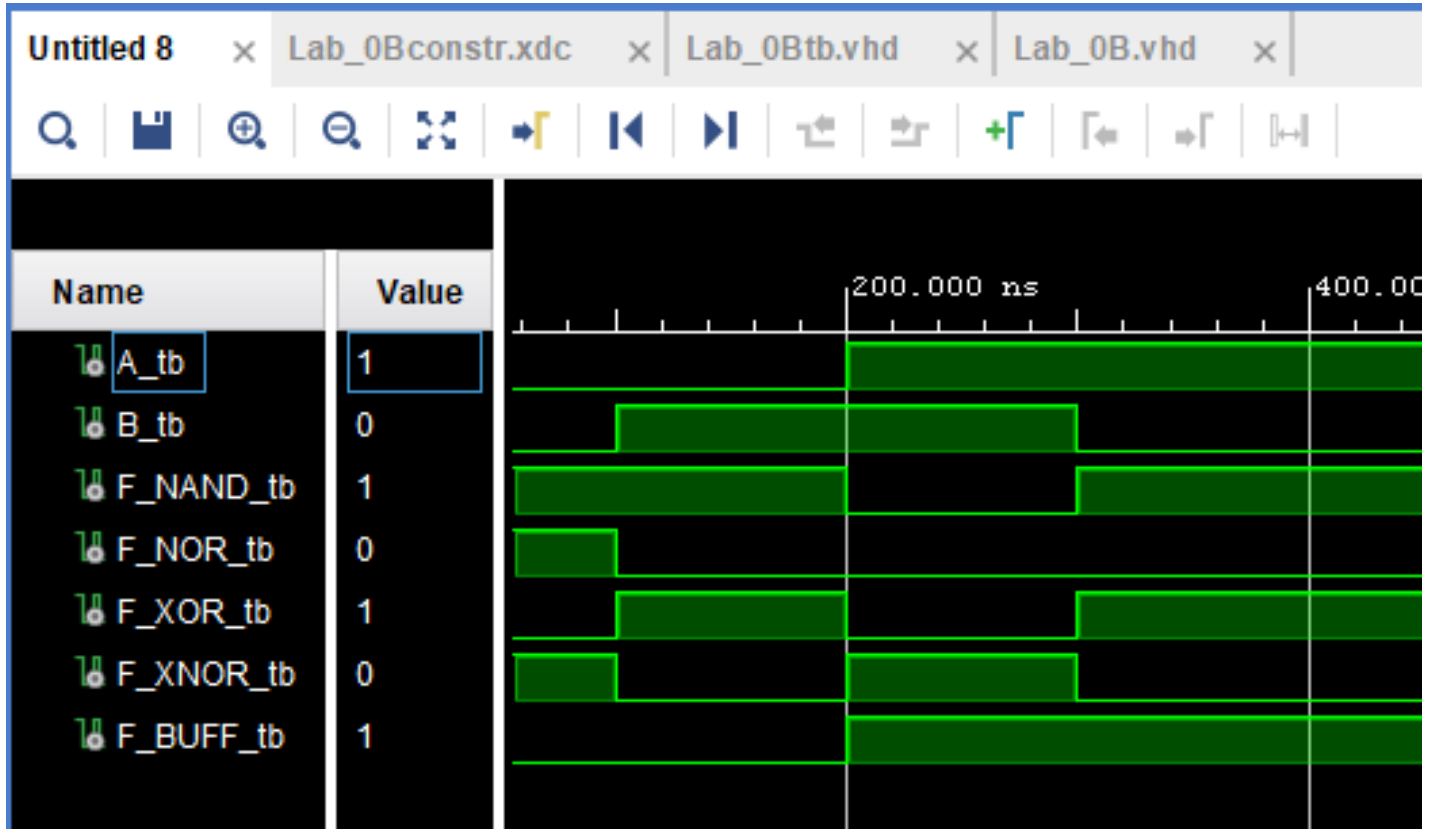
Lab_0Btb.vhd
C:/Users/tsalaz25/ECE238L/Lab_0B/Lab_0B.srcs/sim_1/new/Lab_0Btb.vhd

20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity Lab_0Btb is
35 -- Port ( );
36 end Lab_0Btb;
37
38 architecture bench of Lab_0Btb is
39
40 component Lab_0B
41     Port ( A, B : in STD_LOGIC;
42           F_NAND : out STD_LOGIC;
43           F_NOR : out STD_LOGIC;
44           F_XOR : out STD_LOGIC;
45           F_XNOR : out STD_LOGIC;
46           F_BUFF : out STD_LOGIC);
47 end component;
48
49 signal A_tb, B_tb : STD_LOGIC;
50 signal F_NAND_tb, F_NOR_tb, F_XOR_tb : STD_LOGIC;
51 signal F_XNOR_tb, F_BUFF_tb : STD_LOGIC;
52
53 begin
54
55 uut: Lab_0B port map ( A => A_tb,
56                       B => B_tb,
57                       F_NAND => F_NAND_tb,
58                       F_NOR => F_NOR_tb,
59                       F_XOR => F_XOR_tb,
60                       F_XNOR => F_XNOR_tb,
61                       F_BUFF => F_BUFF_tb);
62
63 stimulus : process
64     begin
65         A_tb <= '0'; B_tb <= '0'; wait for 100 ns;
66         A_tb <= '0'; B_tb <= '1'; wait for 100 ns;
67         A_tb <= '1'; B_tb <= '1'; wait for 100 ns;
68         A_tb <= '1'; B_tb <= '0'; wait for 100 ns;
69         wait;
70     end process;
71
72 end process;

Lab_0Bconstr.xdc
C:/Users/tsalaz25/ECE238L/Lab_0B/Lab_0B.srcs/constrs_1/new/Lab_0Bconstr.xdc

1 set_property PACKAGE_PIN L16 [get_ports A]
2 set_property PACKAGE_PIN J15 [get_ports B]
3 set_property PACKAGE_PIN R18 [get_ports F_BUFF]
4 set_property PACKAGE_PIN H17 [get_ports F_NAND]
5 set_property PACKAGE_PIN K15 [get_ports F_NOR]
6 set_property PACKAGE_PIN J13 [get_ports F_XOR]
7 set_property PACKAGE_PIN N14 [get_ports F_XNOR]
8 set_property IOSTANDARD LVCMOS33 [get_ports A]
9 set_property IOSTANDARD LVCMOS33 [get_ports B]
10 set_property IOSTANDARD LVCMOS33 [get_ports F_BUFF]
11 set_property IOSTANDARD LVCMOS33 [get_ports F_NAND]
12 set_property IOSTANDARD LVCMOS33 [get_ports F_NOR]
13 set_property IOSTANDARD LVCMOS33 [get_ports F_XNOR]
14 set_property IOSTANDARD LVCMOS33 [get_ports F_XOR]
15
```

Wave Form, Truth Table and Schematic



VHDL Source Code, Simulation Code (Test Bench), and Constraints

The image displays three sequential screenshots of a Verilog HDL editor, showing the development of a Verilog module and its testbench.

Left Screenshot (Lab0C_tb.vhd): This is the testbench file. It includes a header with revision information (0.01 - File Created), library declarations for IEEE and UNISIM, and a component declaration for Lab0C. The testbench defines signals A_tb, B_tb, F_NOT_tb, F_AND_tb, and F_OR_tb. It includes a stimulus process that applies test vectors to these signals and a bench procedure that instantiates the Lab0C module.

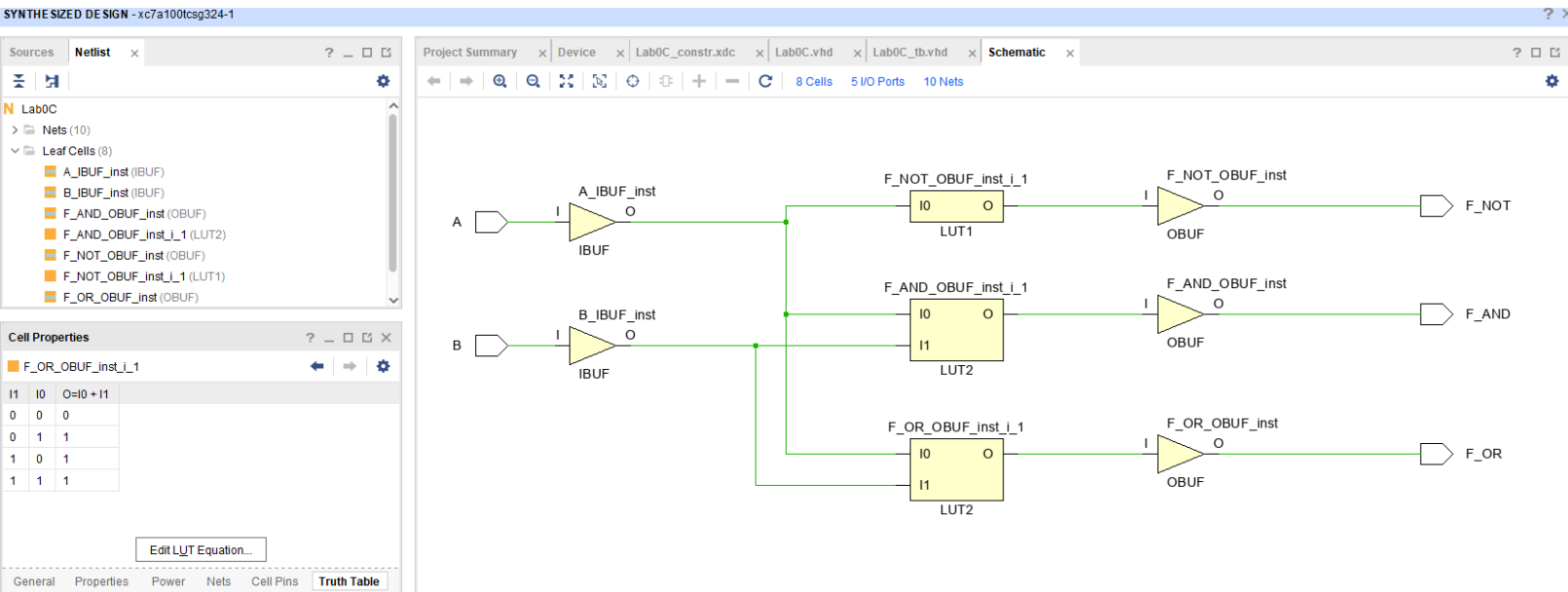
Middle Screenshot (Lab0C.vhd): This is the main module file. It includes a header with revision information (0.01 - File Created), library declarations for IEEE and UNISIM, and a component declaration for Lab0C. The module defines signals A_tb, B_tb, F_NOT_tb, F_AND_tb, and F_OR_tb. It includes a stimulus process that applies test vectors to these signals and a bench procedure that instantiates the Lab0C module.

Right Screenshot (Lab0C_constr.xdc): This is the constraints file. It defines the package pins for the Lab0C module, including IOSTANDARD LVCMOS33, PACKAGE_PIN L16, PACKAGE_PIN J15, PACKAGE_PIN K15, PACKAGE_PIN H17, and PACKAGE_PIN J13.

Wave Form



Truth Table and Schematic



Summary of Work Completed:

Part 0: This was the introduction to the VHDL interface, For this we used 3 inputs [A, B, C] and one output as defined in the VHDL Source Code. The Sim Code was also very simple for this since we are just AND-ing the inputs. The Truth Table and the Waveform reflect the functionality, and this code worked on with the circuit board

Part A: The task is built on the knowledge in the function. We used 2 inputs [A, B] and 3 outputs to reflect the functionality of NOT, AND, and OR. Note that the NOT function only outputs the opposite of A and was not implemented for B. Waveform and Truth table reflect the functionality

Part B: This task was similar to Part A, just with more complex functions, this time NOR, NAND, XNOR, and XOR. We used 2 inputs [A, B] and the outputs were the 4 funtions above, as well as a BUFF (buffer) for A. Truthable and Wave form is more complex but the functionality remained reliable

Part C: This task was the same as Task A, but we used NAND functions only. Using the Truth Table and Waveform for both A and C, we can see that the functionality is the same.

Problems:

The biggest problems that I encountered were using VHDL and the Vivado interface to code in. It was unusually clunky compared to other coding languages. Another minor thing was Syntax and Semantics as VHDL is very explicit and picky.

Advice:

Given that the language is new, it is important to follow the steps precisely and not rush through the learning process. Typing out the code is the best way to learn what is going on. Using the internet to look for what each code file is doing and how they are connected to each other helped me out as the Lab went on. Asking the TA, Matthew, also helps a lot.

Improvements:

The Slides have a lot of information on them. I think that splitting the information up and having a cleaner flow of information can be beneficial for the students. I feel like the clutter on the slides gets messy and is sometimes inconsistent.